Adder Logic for Matrix

a_{IJ}

J →

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{vmatrix}$$

I ↓

$$\begin{vmatrix} a_{11} & a_{21} & a_{31} & a_{41} \\ a_{12} & a_{22} & a_{32} & a_{42} \\ a_{13} & a_{23} & a_{33} & a_{43} \\ a_{14} & a_{24} & a_{34} & a_{44} \end{vmatrix}$$

$I=4, J=4$ $i=3, j=3$ $r=2\ c=2$ $r'=1\ c'=1$ $i=3, j=2$ $r=2\ c=3$ $r'=1\ c'=1$ $i=3, j=1$ $r=2\ c=3$ $r'=1\ c'=2$

$$B_{44} = a_{33}(a_{11}a_{22} - a_{12}a_{21}) + a_{32}(a_{11}a_{23} - a_{13}a_{21}) + a_{31}(a_{12}a_{23} - a_{13}a_{22})$$

$I=4, J=3$ $i=3, j=4$ $r=2\ c=2$ $r'=1\ c'=1$ $i=3, j=2$ $r=2\ c=4$ $r'=1\ c'=1$ $i=3, j=1$ $r=2\ c=4$ $r'=1\ c'=2$

$$B_{34} = (-1)\left[a_{34}(a_{11}a_{22} - a_{12}a_{21}) + a_{32}(a_{11}a_{24} - a_{21}a_{14}) + a_{31}(a_{12}a_{24} - a_{22}a_{14})\right]$$

etc:

Sequence: $I=4, J=4$ $i=3, j=3$ $r=2$ $c=2$ $r'=1$ $c'=1$

3	2	2	3	1	1
3	1	2	3	1	2

$I=4, J=3$ $i=3, j=4$

		2	2	1	1
3	2	2	4	1	1
3	1	2	4	1	2

etc: $I=4, J=2$